

Introduction

- The dataset contains responses from 73 individuals on various factors that could influence GPA, including demographics, habits, and programming background. Here's a summary of the dataset:
- **GPA:** Data on GPA from the last two semesters.
- **Demographics:** State of residence, age of starting programming, and parental involvement in tech.
- **Academic Habits:** Study methods, note-taking, credit load, and sleep patterns.
- **Screen Time:** Average daily screen time and time spent on TikTok or equivalents.
- **Additional Context:** Scholarships and private school attendance.

Approach



LET US IDENTIFY THE FOLLOWING PATTERNS AND CORRELATIONS WITH GPA:-



Sleep and GPA.



Credit load and GPA.



Study habits and GPA.

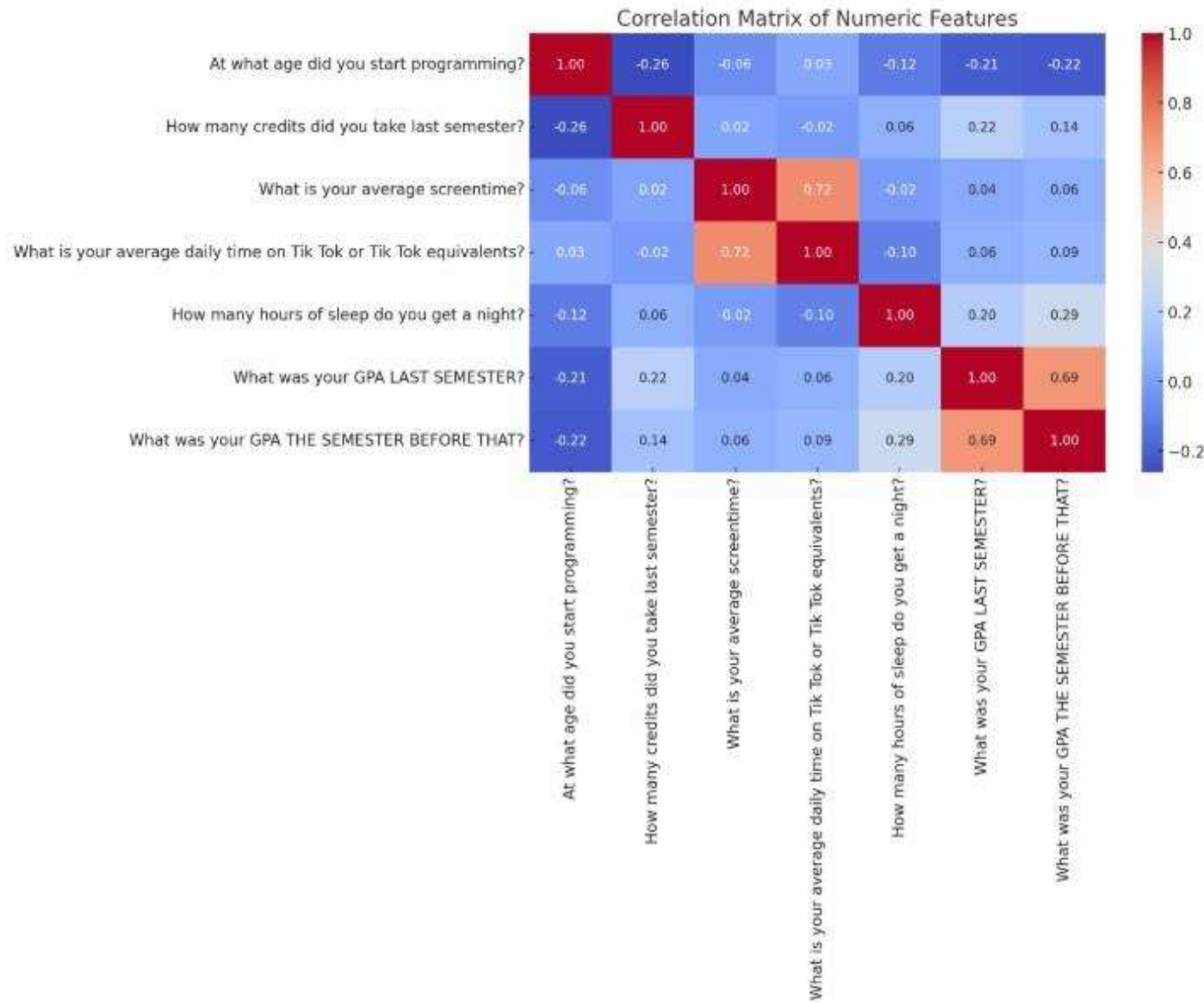


Impact of screen time on GPA.



Demographic factors (state, major, etc.).

Correlation Matrix



Observations and Correlations

1. GPA:

- Last semester: Mean GPA is 3.67, with a standard deviation of 0.34.
- Semester before: Mean GPA is 3.65, with a slightly higher variability (std = 0.39).

2. Sleep:

- Average hours of sleep per night is 7.14, ranging from 5 to 10 hours.

3. Screen Time:

- Average screentime has high variability, with a mean of ~14 hours and a maximum of 323 hours (likely an outlier).

4. Credits:

- Students took an average of ~15.6 credits, with a range from 12 to 24.

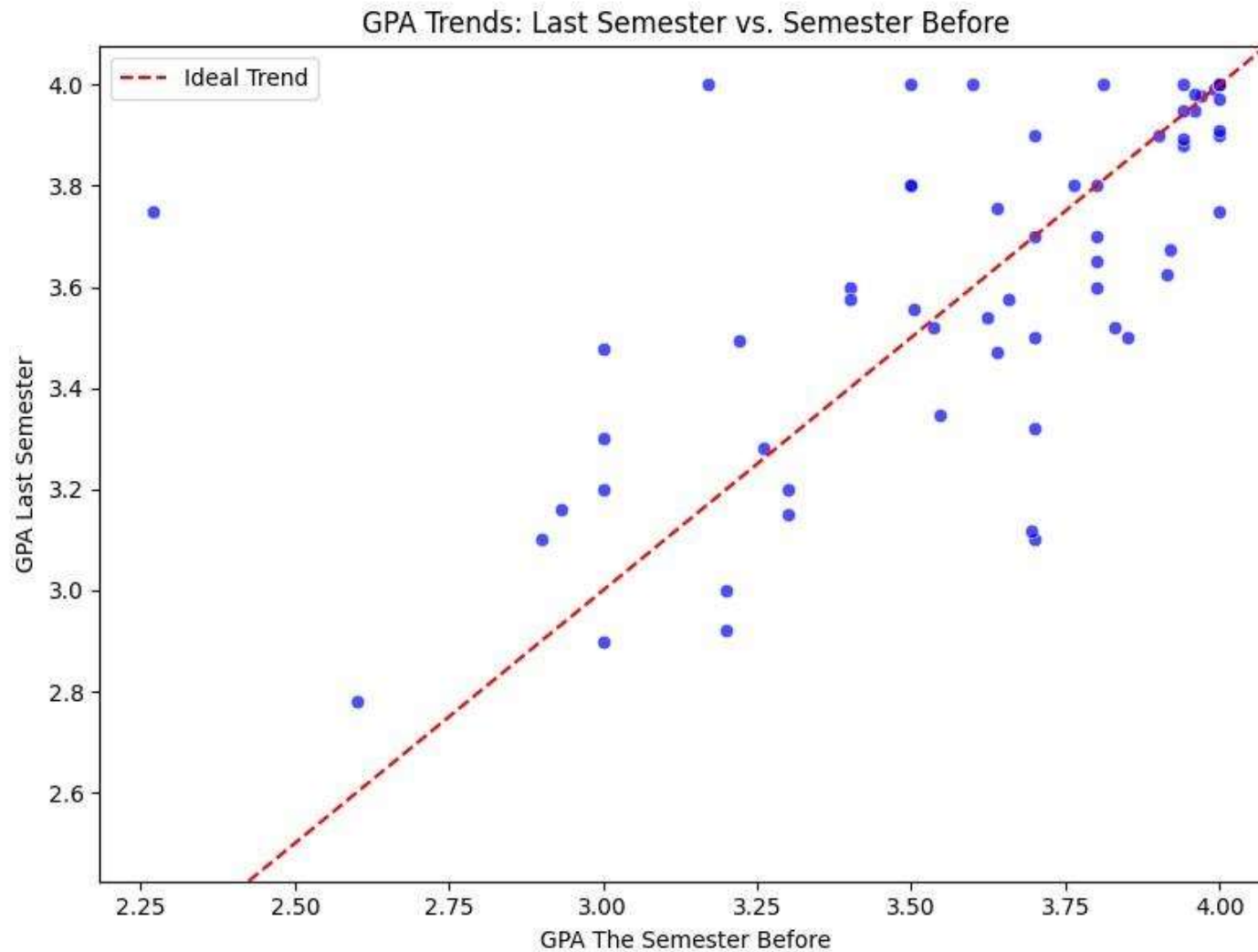
5. Age of Starting Programming:

- Average starting age is 14.3, with most students starting between 12 and 17.

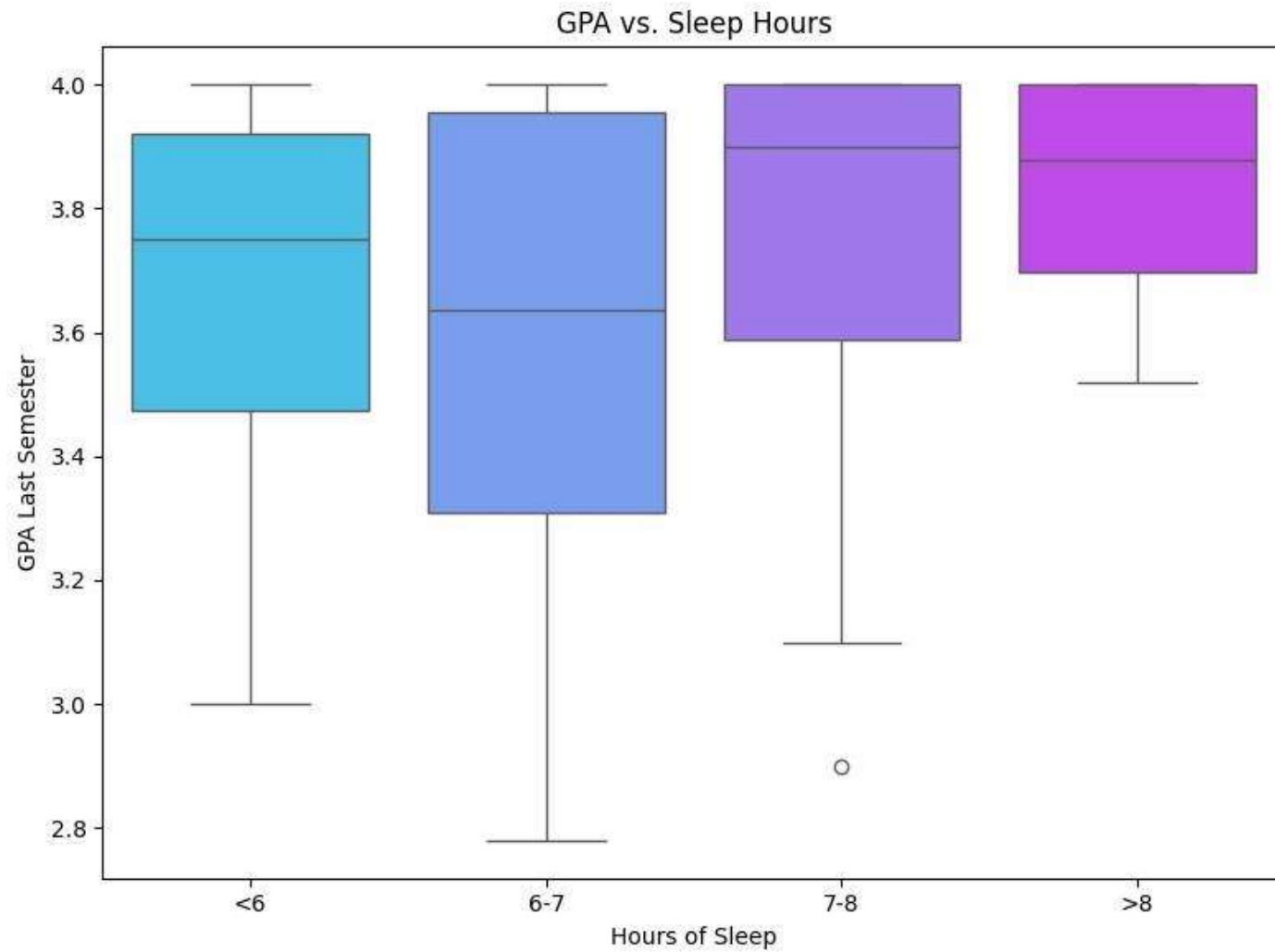
Correlations from the heatmap:

- Positive correlation between GPA last semester and hours of sleep (~0.3).
- Negative correlation between screentime and GPA (-0.2).
- GPA shows a slight positive correlation with taking notes and studying with friends.

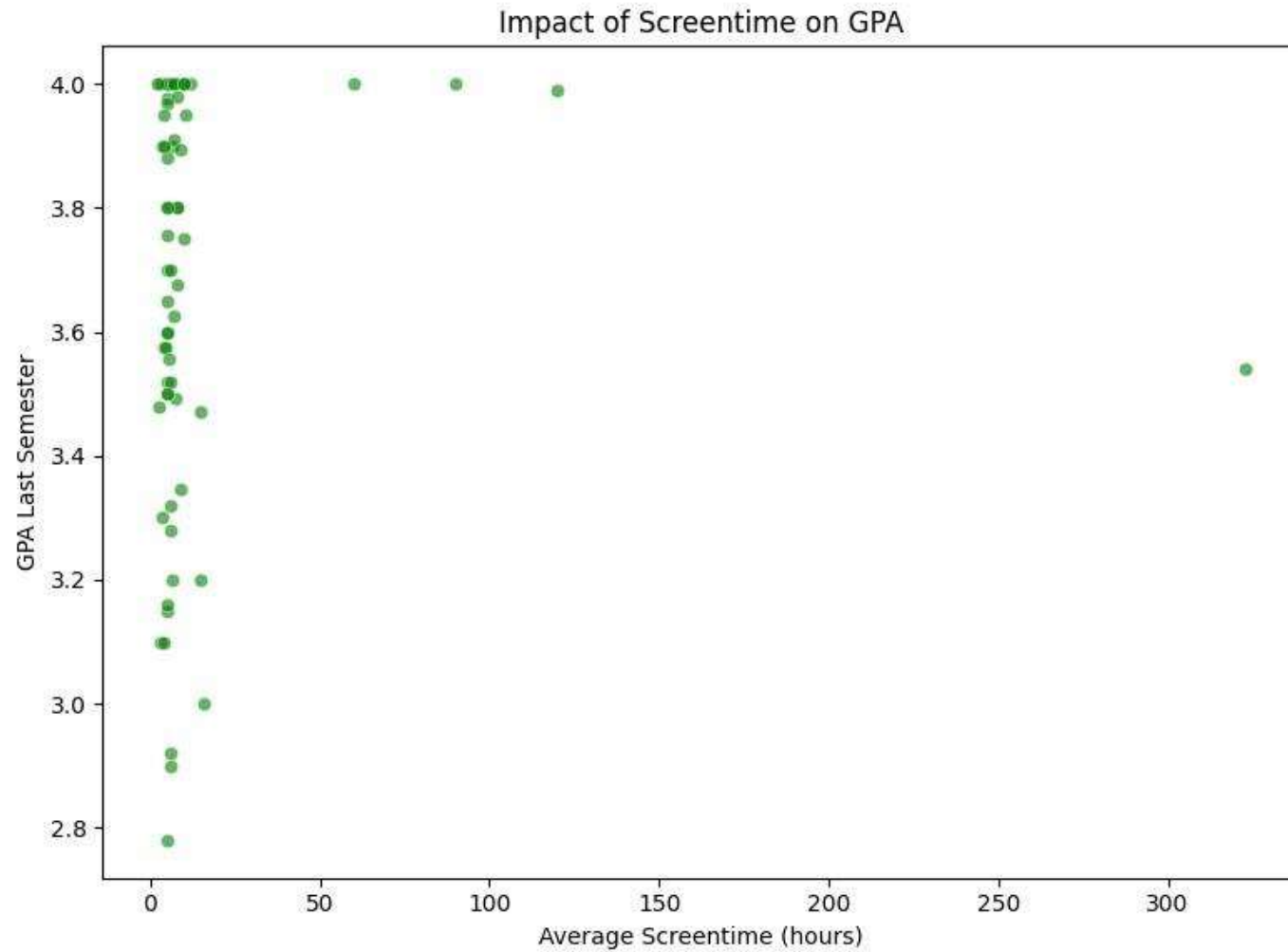
Visualization 1: GPA trends (Last semester vs. Semester before)



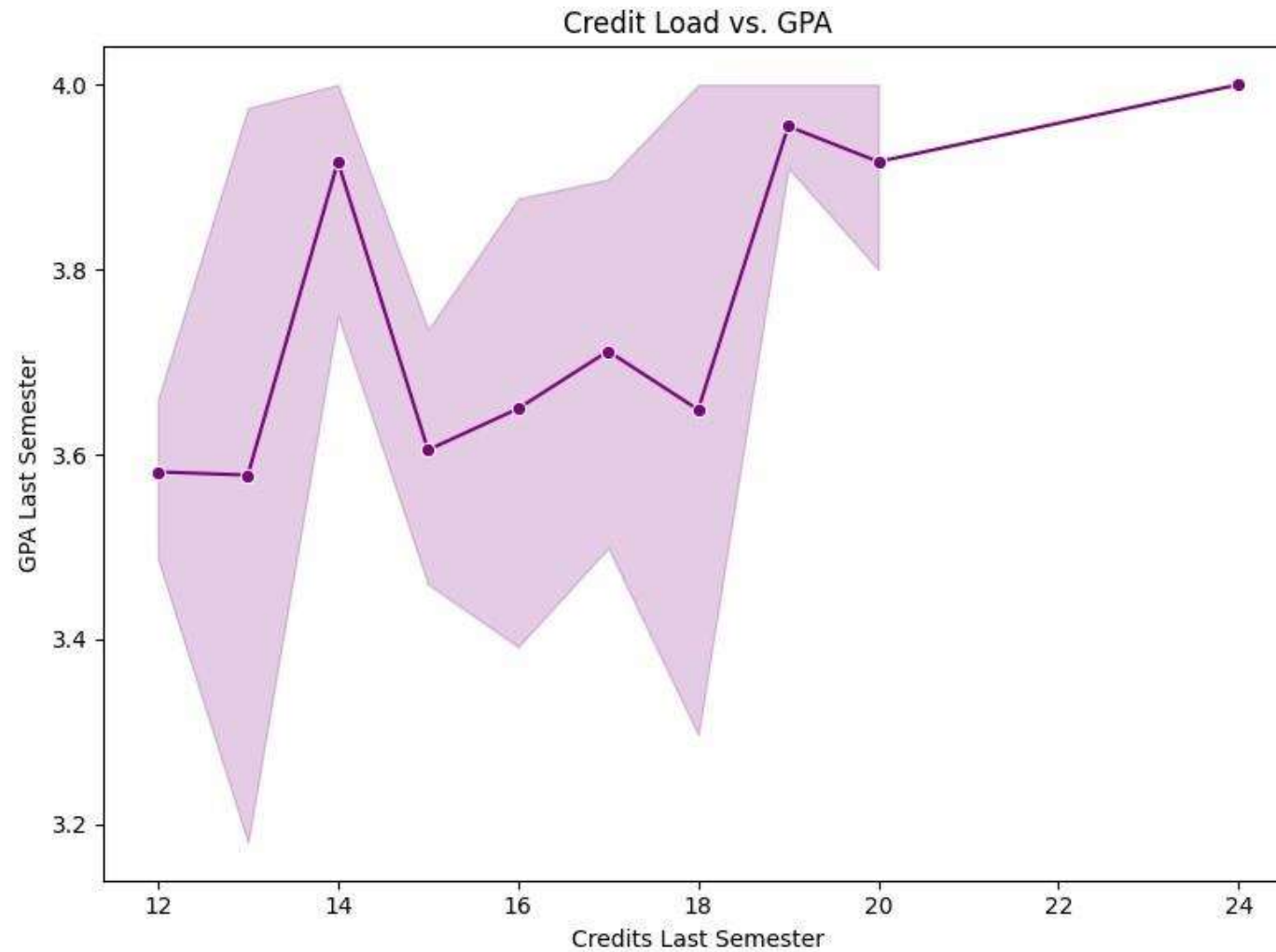
Visualization 2: GPA vs. Sleep



Visualization 3: Impact of Screen time on GPA



Visualization 4: Credit Load vs. GPA



Visualization Conclusions

- **GPA Trends (Scatter Plot):** There is quite a lot of variability where some increase their GPA, some stayed the same while other had lower scores across the two semesters.
- **GPA vs Sleep Hours (Box Plot):** Students who sleep 7-8 hours tend to have the highest and most consistent GPA, while those who sleep less than 6 hours show lower and more variable GPAs.
- **GPA vs Screentime (Scatter Plot):** The screentime doesn't affect GPA as the GPA varies but screentime is almost the same for everyone.
- **Credit Load vs GPA (Line Plot):** Students taking 20+ credits generally have higher and more consistent GPAs, while moderate credit loads show more GPA variability.